

Measure Sinusoidal Voltage in a Circuit

Voltage-divider response observed with the ADALM2000 (M2K) and Scopy Oscilloscope

Circuit and procedure

The same $510\ \Omega$ / $1.1\ \text{k}\Omega$ divider was used. W1 drove the divider input. Channel 1 measured V_{OUT} at the resistor junction relative to GND. For each case, the oscilloscope was run until at least three complete cycles were visible.

Configured input cases

Case	W1 setting at 1 kHz	Input peaks
A	5 Vpp, 0 V offset	-2.5 V to +2.5 V
B	10 Vpp, 0 V offset	-5 V to +5 V
C	5 Vpp, +2.5 V offset	0 V to +5 V

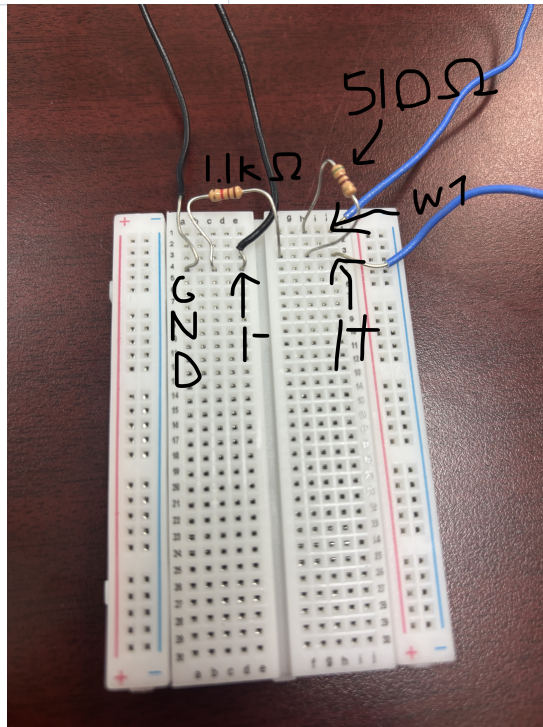


Figure 1. Labeled breadboard photo of the divider used for the sinusoidal measurements.

Oscilloscope evidence

The displayed output waveforms are sinusoidal and show more than three periods in each capture. Output magnitude is reduced by the divider ratio.

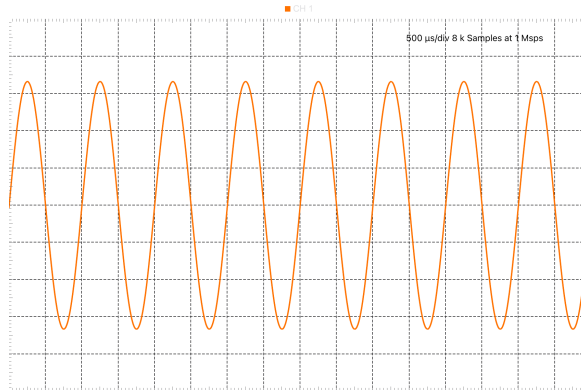


Figure 2. Case A: 5 Vpp, 0 V offset input.

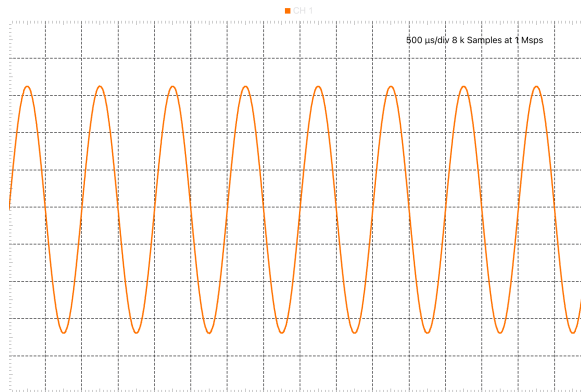


Figure 3. Case B: 10 Vpp, 0 V offset input.

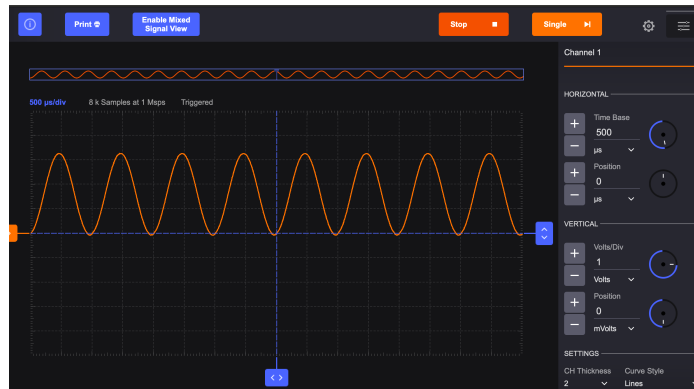


Figure 4. Case C: 5 Vpp, +2.5 V offset input.